

Cloud Computing

What is cloud computing :

Cloud computing means storing and accessing data and programs over the internet instead of your computer's hard drive.

Characteristics of Cloud Computing :

Increase users flexibility

Cost reduction

Device and location independence

Automated management

Virtualization

Types of Cloud Computing Services

Infrastructure-as-a-service(IaaS) : It will give complete empty infrastructure, you can customize by your own specific way

- Used by system admin
- You can choose your own OS
- Provides infrastructure
- Provides IP Address related to it
- Virtual machine and storage

Platform-as-a-service(PaaS) : It will give you a runtime environment by this you can develop or deploy the application by tools which is also provided by vendors.

- Used by developer
- Runtime environment
- No need to purchase expensive hardware or software while making or developing
- Platform where you can work

Software-as-a-service(SaaS) : It will give you a software interface to you for processing all this stuff and backend process will be managed by vendors

- Used by end users or client user
- Servers and resources are managed by the vendors

- No need to installing software, you will require only good internet connection
- Platform independency doesn't require any specific platform
- No access to all the resources

Cloud Deployment Models

Cloud deployment model selection is based on the enterprise requirements.

Public cloud : Services are rendered over a network that is open for public

- Less cost
- Globally available
- Less customization

Private cloud : Cloud infrastructure operated solely for a single organization

- High cost
- Locally available
- High customization

Hybrid cloud : Composition of public and private cloud

- Cost efficient
- High customization
- Less cost

What is virtualization ?

It is the ability to run multiple operating system on a simple physical system and share the underlying resources such as a server, a storage device or a network

Benefits from Virtualization :

- Increase business continuity
- Reduce setup cost
- Facilities better backup and data protections

Security concerns of Cloud

1. Third party handling data



2. Challenging cyber attacks

3. Insider threats

4. Government intrusion

5. Lack of support

Cloud Computing attacks

- Service hijacking using social Engineering attacks
- Service hijacking using network sniffing
- Session hijacking using XSS attack
- Domain name system attack
- SQL injection attacks
- DOS and DDoS
- Man in the cloud attack
- Insecure encryption and many more